

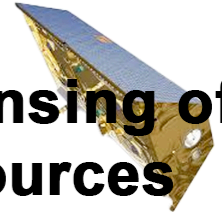


Australian
National
University

Research School
of Earth Sciences



**Remote Sensing of
Water Resources**
EMSC3025/6025



Global Water Cycle Overview

EMSC3025/6025 — Recap Session

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EMSC3025/6025: Remote Sensing of Water
Resources

Dr. Sia Ghelichkhan

Objectives of this Recap Session

By the end of this session, you should be able to:

- **Synthesise** the fundamental processes of the hydrological cycle
- **Apply** the water balance equation to different environments
- **Explain** precipitation formation and measurement challenges
- **Evaluate** evaporation controls and estimation methods
- **Analyse** runoff generation mechanisms and streamflow measurement
- **Integrate** these processes at catchment and global scales

This recap consolidates the foundation modules of the water course.

The Big Picture: Global Water Distribution

- Earth's water is highly concentrated:
 - Oceans/seas: **96.5%**
 - Ice/glaciers: **1.74%**
 - Groundwater: **1.69%**
 - Accessible freshwater: **~0.27%**

The challenge: Uneven distribution in space and time

Storage	Volume ($\times 10^3 \text{ km}^3$)	% of total
Oceans/seas	1,338,000	96.54
Ice caps/glaciers	24,064	1.74
Groundwater	23,400	1.69
Lakes	176	0.013
Atmosphere	12.9	0.0009
Rivers	2.12	0.00015

Table 1: Global water distribution - the foundation of all hydrology

The Water Balance Equation: Our Central Tool

The **water balance equation** represents mass conservation:

$$P \pm E \pm \Delta S \pm Q = 0$$

Often rearranged as: $Q = P - E - \Delta S$

Where:

- P : Precipitation (input)
- E : Evaporation (output)
- ΔS : Change in storage
- Q : Runoff/discharge (output)

Applications vary by environment:

- **Humid regions:** P dominates
- **Arid regions:** E dominates
- **Cold regions:** ΔS (snow/ice) critical
- **Urban areas:** Modified by infrastructure

Catchments: Our Spatial Framework

- A **catchment** is defined by **topographic divides**
- Surface water vs groundwater boundaries may differ
- **Nested hierarchy**: sub-catchments within larger basins

Manual delineation steps:

1. Identify outlet point
2. Trace flow lines (perpendicular to contours)
3. Identify ridgelines
4. Connect divides to form boundary
5. Verify all flow drains to outlet

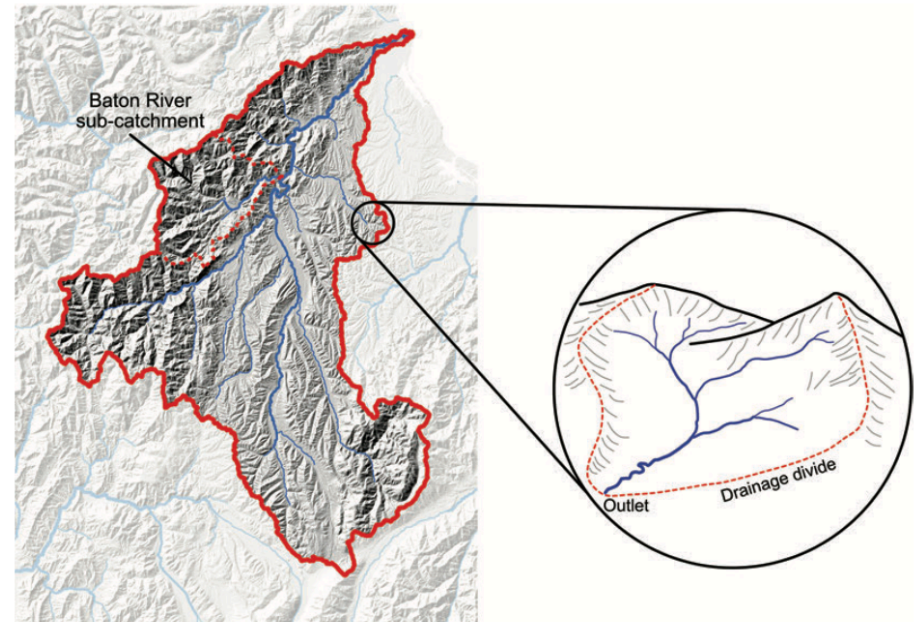


Figure 1: Catchment boundaries define our study units

Precipitation: Water Input to the System

Precipitation Formation: Three Essential Conditions

For precipitation to occur:

1. Cooling of the atmosphere

- Adiabatic cooling via uplift
- DALR: $\sim 9.8^{\circ}\text{C}/\text{km}$ (dry air)
- SALR: $\sim 4\text{--}6^{\circ}\text{C}/\text{km}$ (saturated air)

2. Condensation nuclei

- Dust, smoke, sea salt particles
- Enable droplet formation

3. Droplet growth

- From $\sim 1\mu\text{m}$ to $\sim 3000\mu\text{m}$
- Via collision-coalescence or Bergeron process

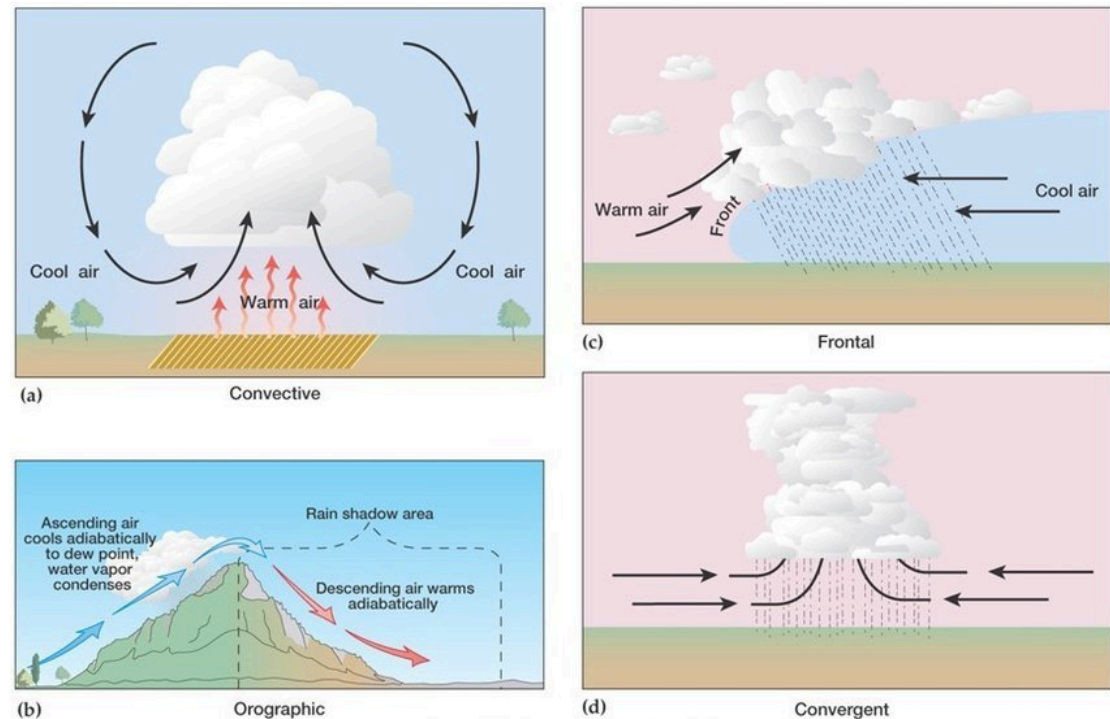


Figure 2: Three main lifting mechanisms

Precipitation Distribution Controls

Static influences (fixed landscape features):

- **Altitude:** Higher elevations → more rainfall
- **Aspect:** Windward slopes → more precipitation
- **Slope:** Minor effect at small scales

Dynamic influences (atmospheric processes):

- **Storm tracks:** Cyclone paths
- **Frontal systems:** Air mass interactions
- **Seasonal patterns:** Monsoons, ITCZ shifts

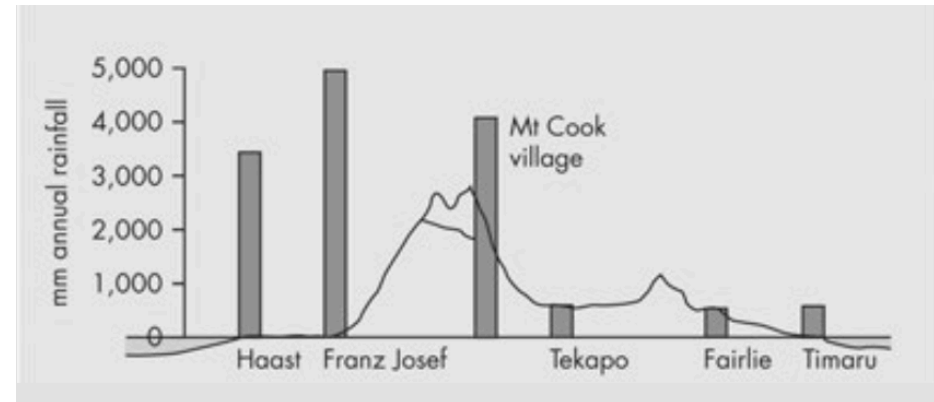


Figure 3: Rain shadow effect in New Zealand

Precipitation Measurement Challenges

Four major error sources:

1. **Evaporation:** Water lost before measurement
2. **Wetting:** Water adheres to funnel walls
3. **Rain splash:** Water splashes in/out
4. **Wind turbulence:** Reduces catch efficiency

Solutions:

- Funnel design minimizes evaporation
- Steep angles reduce wetting
- Wind shields reduce turbulence
- Non-splash grids prevent splash errors

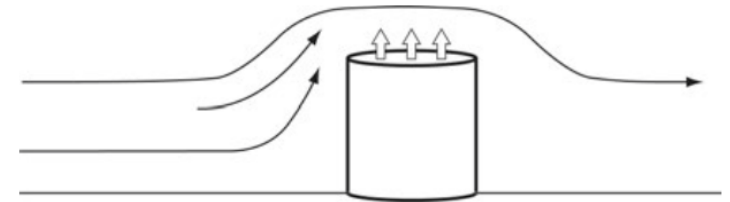


Figure 4: Wind effects on rain gauge accuracy

From Point to Areal Precipitation

Challenge: Convert point measurements to catchment-wide estimates

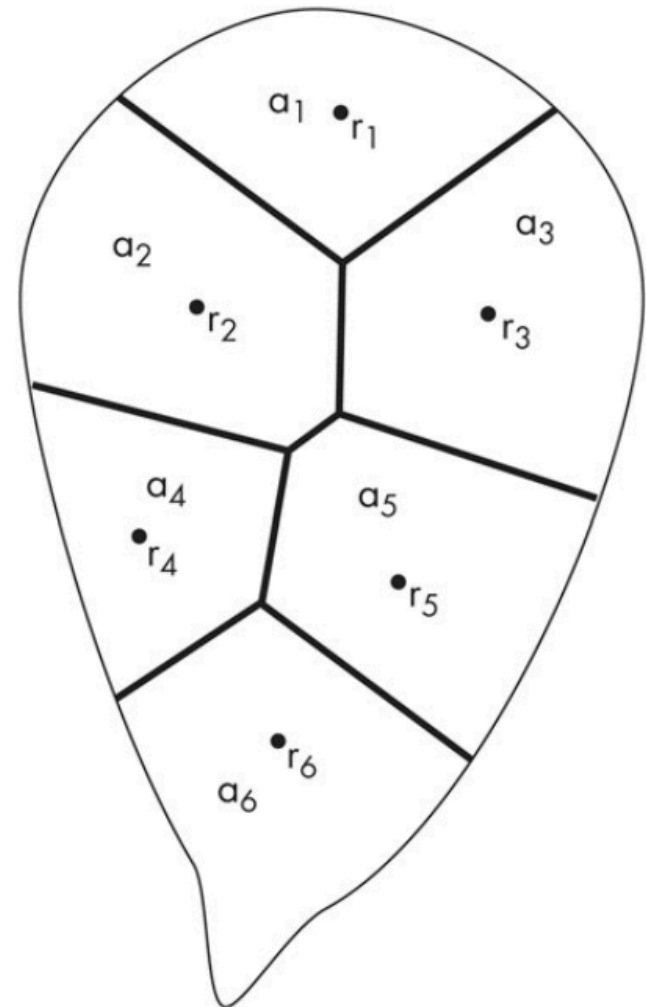
Three main methods:

1. **Thiessen polygons:** Area-weighted average $R =$

$$\sum_{i=1}^n r_i \times \left(\frac{a_i}{A} \right)$$

2. **Hypsometric method:** Elevation-based

3. **Isohyetal method:** Contour-based interpolation



Evaporation: Water Output from the System

Types of Evaporation

Three key definitions:

- **Open water evaporation (E_O):** From lakes, rivers, oceans
- **Potential evaporation (E_{pt}):** Maximum possible with unlimited water
- **Actual evaporation (E_t):** What actually occurs

Components:

- **Soil evaporation:** Direct from soil matrix
- **Transpiration:** Through plant stomata
- **Interception loss:** From wet vegetation surfaces

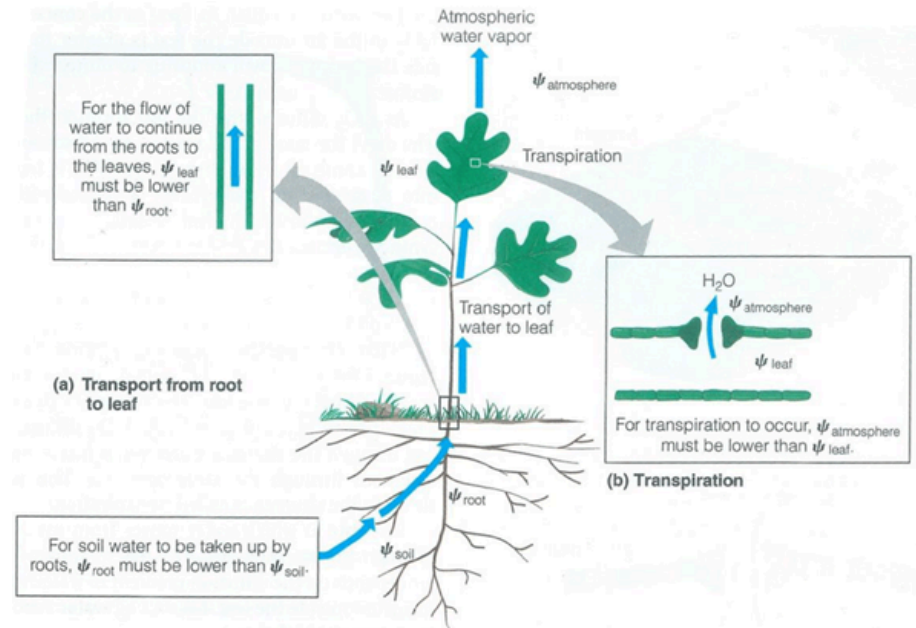


Figure 6: Components of evapotranspiration

Controls on Evaporation

Three fundamental controls:

1. Available energy (Q^*): $Q^* = Q_S \pm Q_L \pm Q_G$

- Solar radiation dominates
- Net radiation drives the process

2. Water availability:

- Unlimited over oceans
- Limited by soil moisture on land
- Plant access via root systems

3. Atmospheric demand:

- Vapour pressure deficit (VPD)
- $VPD = e_s - e_a$
- Higher VPD $\rightarrow$ greater evaporation potential

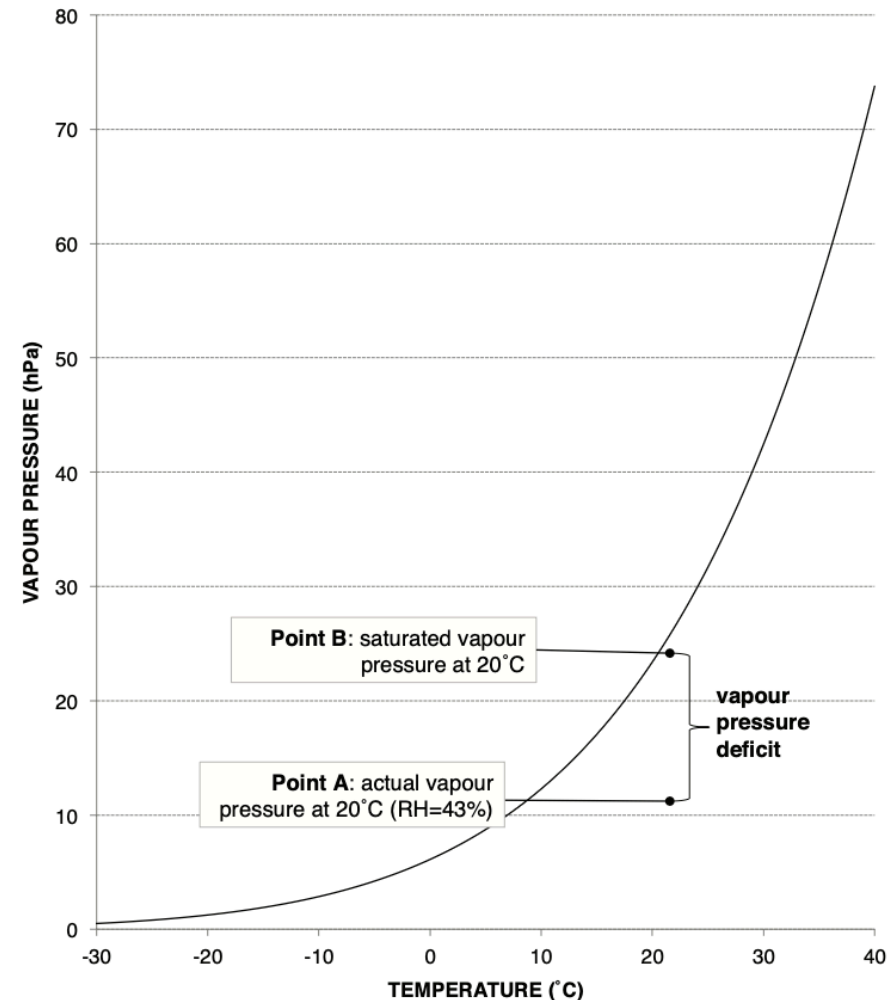


Figure 7: Vapour pressure deficit controls atmospheric demand

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Measuring Evaporation: The Challenge

Direct methods:

- **Eddy covariance:** Most accurate, expensive
- **Bowen ratio:** Uses temperature/humidity gradients
- **Aerodynamic profile:** Flux-gradient relationships

Indirect methods:

- **Evaporation pans:** Simple but overestimate
- **Lysimeters:** Most accurate for vegetated surfaces
- **Water balance:** $E = \Delta S - P - Q$

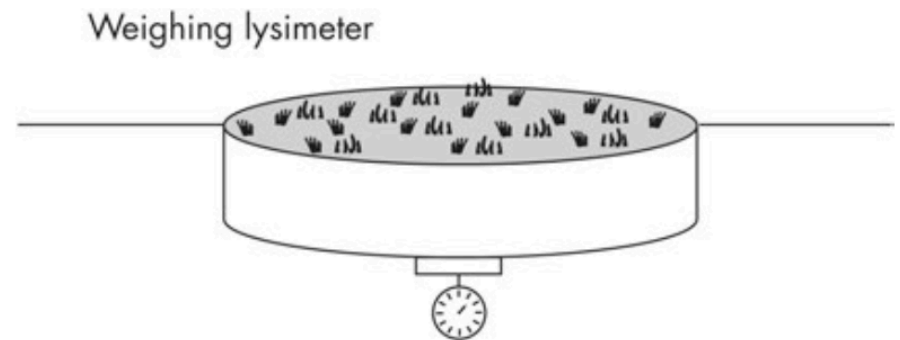


Figure 8: Weighing lysimeter for accurate ET measurement

Evaporation Estimation Methods

Thornthwaite method (temperature-based):

$$E_p = 16b \left(\frac{10T}{I} \right)^a$$

- Simple, requires only temperature
- Monthly scale only
- Regional calibration needed

Penman equation (combination method):

$$E_o = \frac{\Delta Q^* + \gamma E_a}{\Delta + \gamma}$$

- Combines energy balance and aerodynamic terms
- More accurate, requires more data

Relationship between actual and potential:

- Depends on water availability
- Non-linear relationship with soil moisture
- Vegetation adds stomatal control

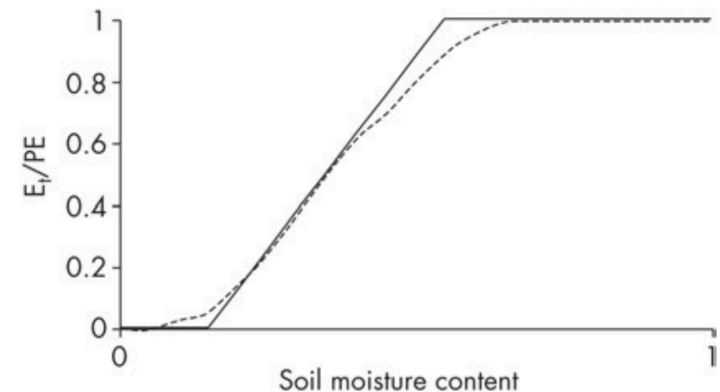


Figure 9: Actual vs potential evaporation relationship

Runoff: Water Movement to Channels

Understanding Hydrographs

Components of a storm hydrograph:

- **Rising limb:** Steep increase to peak
- **Peak discharge:** Maximum flow rate
- **Recession limb:** Gradual decline

Controls on shape:

- Catchment size, slope, shape
- Soil characteristics
- Vegetation cover
- Antecedent moisture conditions

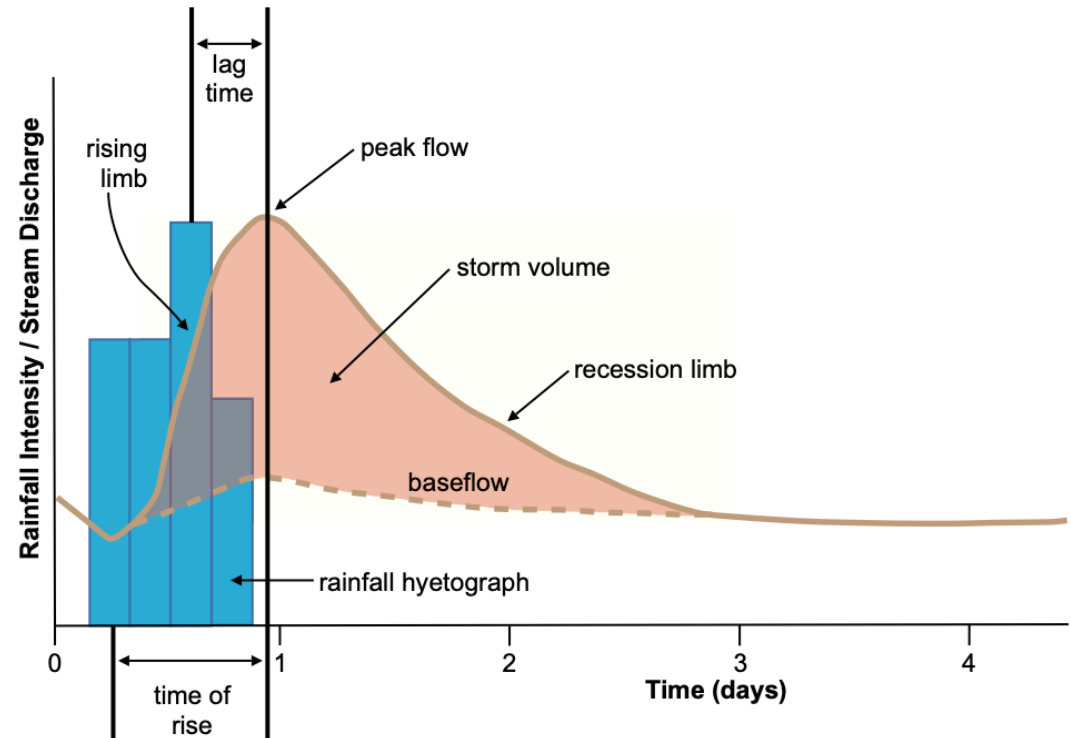


Figure 10: Typical storm hydrograph components

Runoff Generation Theories

Three key theories:

1. Horton (1933): Infiltration-excess overland flow

- Occurs when rainfall > infiltration capacity
- Uniform across catchment

2. Betson (1964): Partial contributing areas

- Only specific zones produce overland flow
- Retains infiltration control

3. Hewlett & Hibbert (1967): Saturation overland flow

- Water table rises to surface
- Variable source areas near channels

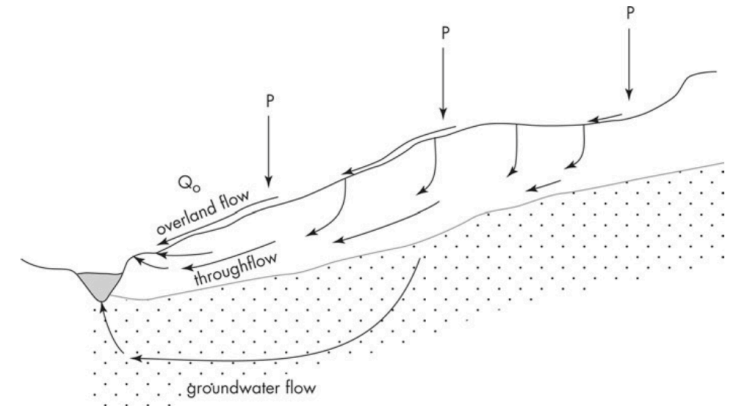


Figure 11: Hillslope runoff processes

The Modern View: Variable Source Areas

Key insights:

- All theories capture aspects of truth
- **Saturated overland flow** dominates in humid regions
- **Variable source area** concept best represents reality
- Contributing areas expand/contract with wetness

Subsurface contributions:

- **Throughflow**: Lateral flow through soil
- **Piston flow**: New water displaces old water
- **Macropores**: Rapid preferential flow paths

Theory	Mechanism	Environment
Horton	Infiltration-excess	Arid/urban
Betson	Partial areas	Semi-arid
Hewlett & Hibbert	Saturation	Humid

Table 2: Runoff theories and their applications

Groundwater's Rapid Role

Capillary fringe hypothesis (Farvolden, 1979):

- Groundwater ridge forms near streams during recharge
- Infiltration saturates unsaturated zone
- Creates hydraulic gradients driving discharge
- **Non-linear** water-saturation relationships

Evidence from tracer studies:

- Significant “old water” in storm runoff
- Rapid groundwater contributions confirmed
- Piston flow and macropore evidence

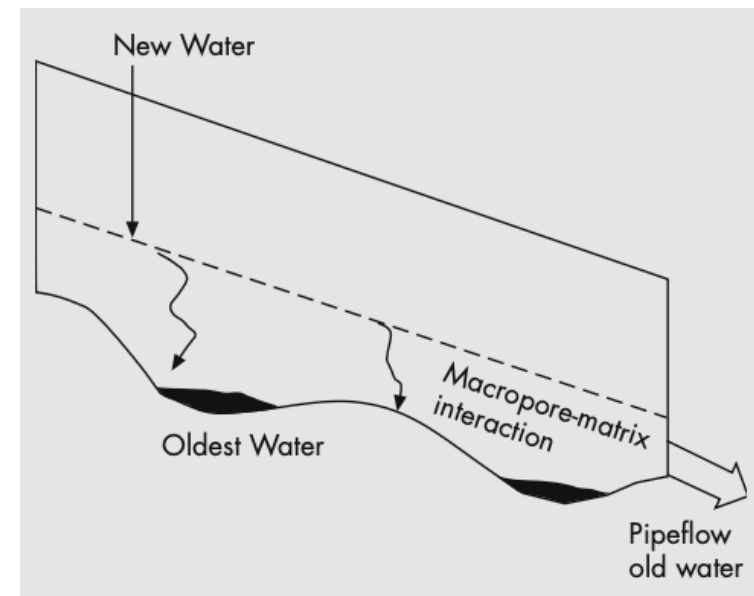


Figure 12: Groundwater contributions at Maimai

Measuring Streamflow

Velocity-area method:

- $Q = V \times A$
- Measure velocity at 60% depth
- Good for small rivers
- Challenges in turbid/shallow streams

Stage-discharge relationships:

- Develop rating curves: stage vs discharge
- Enable continuous monitoring
- Assume stable channel geometry
- Errors increase at high flows

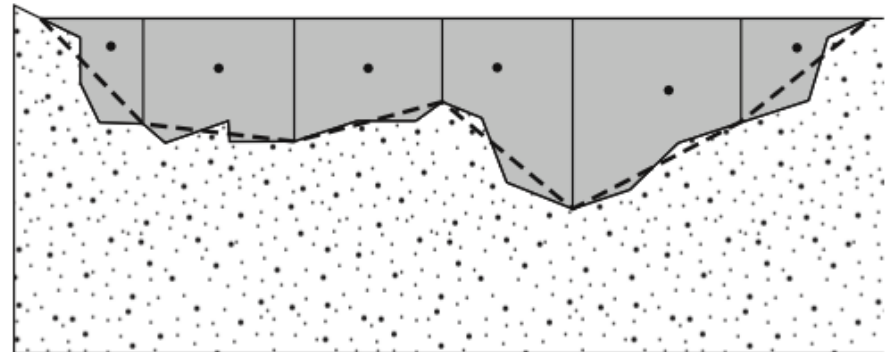


Figure 13: Velocity-area measurement method

Gauging Structures: Weirs and Flumes

Advantages:

- Standardize flow through known geometry
- Enable continuous discharge records
- Known discharge equations

V-notch weir equation: $Q = 1.366 h^{2.5}$ (for 90° notch)

Design considerations:

- V-notches excellent for low flows
- Flumes reduce sedimentation
- Must handle peak flows without bypass

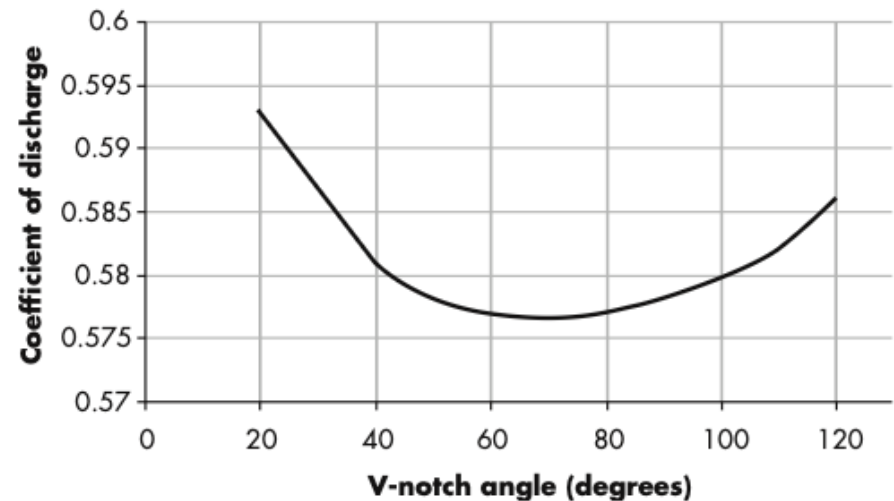


Figure 14: V-notch weir discharge coefficients

Integration: Putting It All Together

The water balance in practice:

- **Precipitation:** Highly variable input
- **Evaporation:** Energy and water limited
- **Runoff:** Complex generation mechanisms
- **Storage:** Critical buffer, hard to measure

Scale considerations:

- **Point measurements vs areal estimates**
- **Process understanding vs practical application**
- **Data availability vs model complexity**

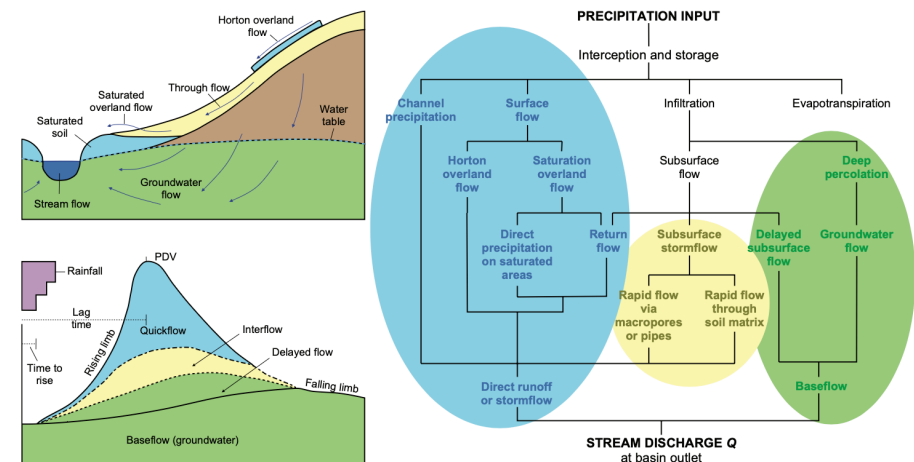


Figure 15: Integrated runoff processes

Climate and Environmental Controls

How climate shapes the water balance:

Humid temperate regions:

- Balanced P, E, Q components
- Saturated overland flow dominates
- Groundwater contributions significant

Arid/semi-arid regions:

- High E/P ratios
- Flash flooding from intense rainfall
- Storage changes critical

Cold regions:

- Snow storage delays runoff
- Seasonal melt dominates discharge
- Frozen ground affects infiltration

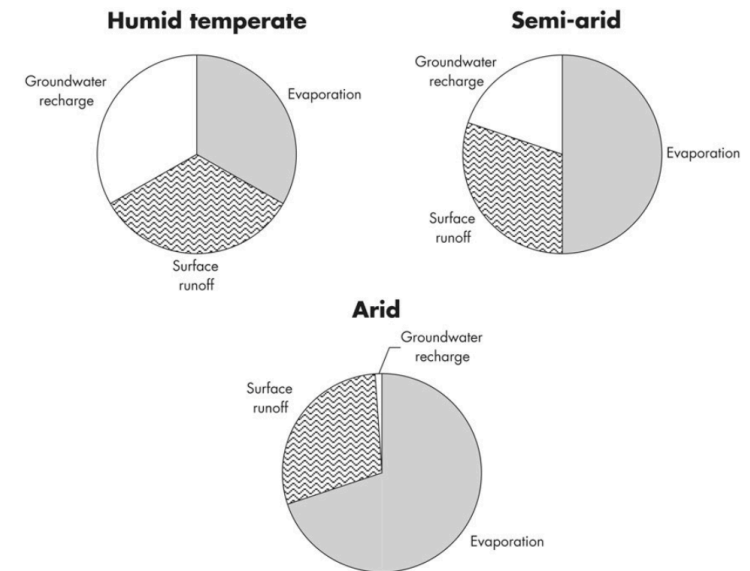


Figure 16: Climate zone water partitioning

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Measurement and Estimation Challenges

Common challenges across all components:

Spatial variability:

- Point measurements vs areal needs
- Scaling issues
- Representative sampling

Temporal variability:

- Event-based vs continuous processes
- Data resolution mismatches
- Long-term vs short-term patterns

Uncertainty sources:

- Instrument errors and calibration
- Environmental factors (wind, temperature)
- Model assumptions and parameters
- Data gaps and quality issues

Solutions:

- Multiple measurement methods
- Quality control procedures
- Uncertainty quantification
- Model validation

Practical Applications: Tutorial Insights

Key problem-solving approaches:

Water balance calculations:

- Identify dominant terms for different environments
- Account for measurement uncertainties
- Consider temporal and spatial scales

Precipitation analysis:

- Understand formation mechanisms
- Evaluate measurement errors
- Apply spatial estimation methods

Evaporation estimation:

- Choose appropriate method for data availability
- Understand limitations and assumptions
- Consider vegetation and soil controls

Runoff analysis:

- Interpret hydrograph characteristics
- Identify dominant generation mechanisms
- Apply measurement techniques appropriately

Case Study Integration: Real-World Examples

Examples from the course:

New Zealand rain shadow:

- Orographic precipitation effects
- Windward vs leeward contrasts
- Measurement challenges in mountains

Darwin thunderstorms:

- Convective precipitation formation
- Tropical climate characteristics
- Measurement in high-intensity events

Corn sweat phenomenon:

- Transpiration impacts on local climate
- Agricultural water use
- Regional-scale evapotranspiration

Maimai experiments:

- Groundwater contributions to streamflow
- Tracer study insights
- Process understanding vs traditional theory

Summary: The Global Water Cycle

Key takeaways from this overview:

- The **water balance equation** provides our fundamental framework
- **Precipitation** formation requires specific atmospheric conditions and faces measurement challenges
- **Evaporation** is controlled by energy, water availability, and atmospheric demand
- **Runoff** generation involves complex interactions between surface and subsurface processes
- **Integration** across scales and processes is essential for understanding catchment hydrology
- **Measurement and estimation** methods each have strengths, limitations, and appropriate applications

Questions for Review and Discussion

Conceptual questions:

- How do the three evaporation types (E_0 , E_{pt} , E_t) relate to each other?
- Why is the variable source area concept considered superior to earlier runoff theories?
- How do measurement errors compound when estimating catchment water balance?

Application questions:

- Given climate data, which evaporation estimation method would you choose and why?
- How would you design a precipitation monitoring network for a mountainous catchment?
- What factors would influence your choice of streamflow measurement method?

Ready to apply these concepts to real-world hydrological problems!